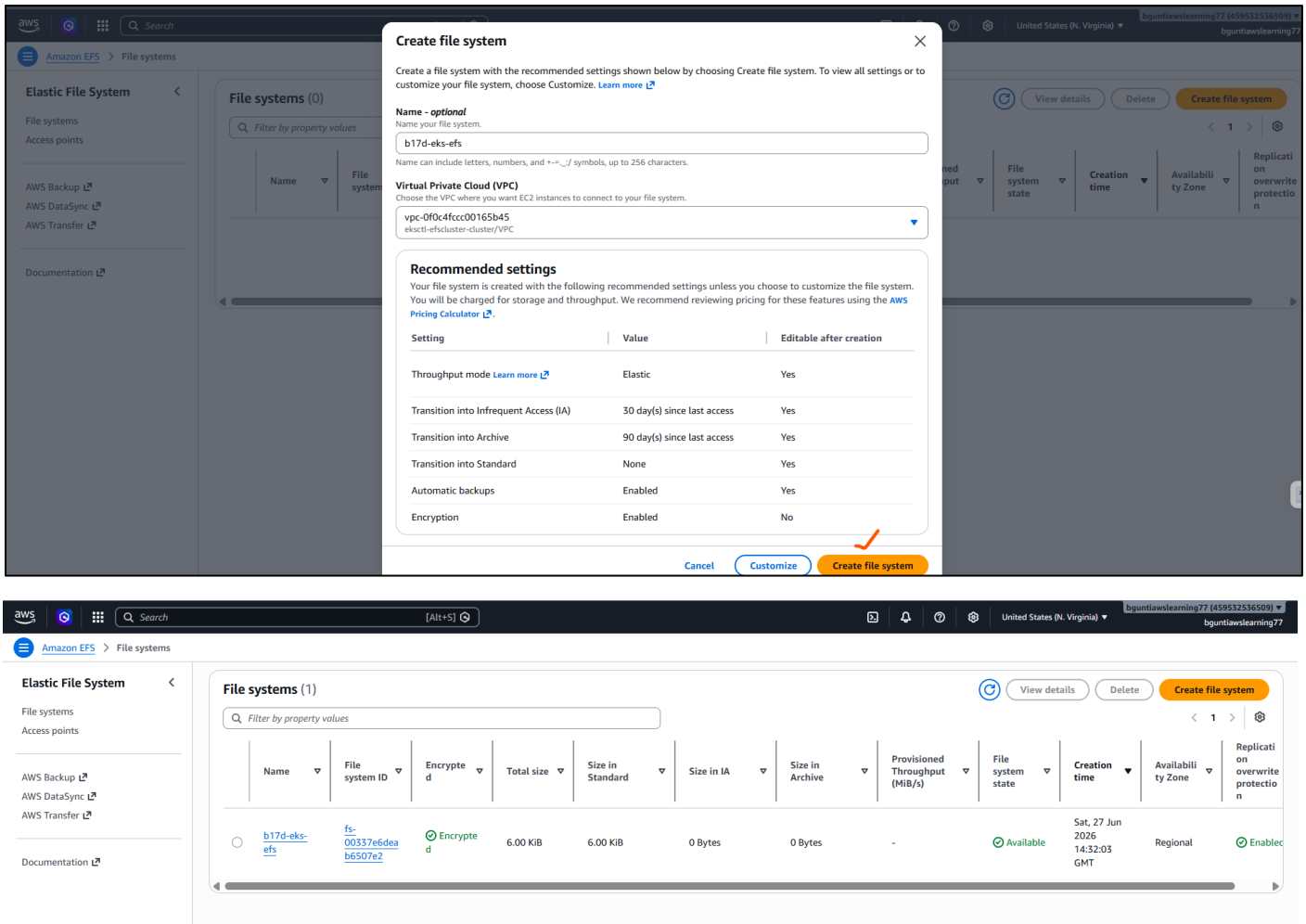


Kubernetes EFS Volume provisioning

Amazon Elastic File System - Scalable, elastic, cloud-native NFS file system



```
kubectl apply -k github.com/kubernetes-sigs/aws-efs-csi-driver/deploy/kubernetes/overlays/stable/ecr/?ref=release-3.X
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ eksctl create cluster --name efsccluster --nodegroup-name b17dng --node-type m7i-flex.large --nodes 3 --managed
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ eksctl get cluster
```

```
NAME          REGION    EKSTL CREATED
efsccluster   us-east-1 True
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ kubectl get nodes
```

```
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-18-202.ec2.internal      Ready    <none>    59m    v1.34.9-eks-93b80c6
ip-192-168-22-87.ec2.internal       Ready    <none>    59m    v1.34.9-eks-93b80c6
ip-192-168-41-115.ec2.internal      Ready    <none>    59m    v1.34.9-eks-93b80c6
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ kubectl apply -k github.com/kubernetes-sigs/aws-efs-csi-driver/deploy/kubernetes/overlays/stable/ecr/?ref=release-3.X
error: failed to run 'C:\Program Files\Git\mingw64\bin\git.exe fetch --depth=1 https://github.com/kubernetes-sigs/aws-efs-csi-driver release-3.X': fatal: couldn't find remote ref release-3.X
: exit status 128
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/kubernetes (main)
$ kubectl apply -k github.com/kubernetes-sigs/aws-efs-csi-driver/deploy/kubernetes/overlays/stable/ecr/?ref=release-2.1
# warning: 'bases' is deprecated. Please use 'resources' instead. Run 'kustomize edit fix' to
update your kustomization automatically.
serviceaccount/efs-csi-controller-sa created
serviceaccount/efs-csi-node-sa created
clusterrole.rbac.authorization.k8s.io/efs-csi-external-provisioner-role created
clusterrole.rbac.authorization.k8s.io/efs-csi-external-provisioner-role-describe-secrets created
clusterrole.rbac.authorization.k8s.io/efs-csi-node-role created
rolebinding.rbac.authorization.k8s.io/efs-csi-provisioner-binding-describe-secrets created
clusterrolebinding.rbac.authorization.k8s.io/efs-csi-node-binding created
clusterrolebinding.rbac.authorization.k8s.io/efs-csi-provisioner-binding created
deployment.apps/efs-csi-controller created
daemonset.apps/efs-csi-node created
csidriver.storage.k8s.io/efs.csi.aws.com configured
```

The screenshot shows the AWS Management Console with the 'Instances' page selected. A table lists three EC2 instances, all in a 'Running' state. The first instance, 'efscluster-b17dng-Node' with ID 'i-089a5ce9cab3d4a70', is highlighted. Below the table, the 'Security' tab for this instance is open, showing its IAM role 'eksctl-efscluster-nodegroup-b17dng-NodeInstanceRole-RQ0amXZ2lbaO' and its security group 'sg-0a0bbb73d748b307d'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
efscluster-b17dng-Node	i-089a5ce9cab3d4a70	Running	m7i-flex.large	3/3 checks passed	us-east-1a	ec2-44-204-165-232.co...	44.204.165.232	-
efscluster-b17dng-Node	i-0f360d17d7e0dcf92	Running	m7i-flex.large	3/3 checks passed	us-east-1a	ec2-44-213-99-30.com...	44.213.99.30	-
efscluster-b17dng-Node	i-0f2712848b506fd53	Running	m7i-flex.large	3/3 checks passed	us-east-1f	ec2-44-197-103-91.co...	44.197.103.91	-

The screenshot shows the AWS IAM console for the role 'eksctl-efscluster-nodegroup-b17dng-NodeInstanceRole-RQ0amXZ2lbaO'. The 'Permissions policies' tab is active, displaying a list of four managed policies: 'AmazonEC2ContainerRegistryPullOnly', 'AmazonEKS_CNI_Policy', 'AmazonEKSEKSWorkerNodePolicy', and 'AmazonSSMManagedInstanceCore'. A red checkmark is placed over the 'Add permissions' button, which has a dropdown menu showing 'Attach policies' and 'Create inline policy'.

The screenshot shows the 'Add Permissions' dialog box. It lists 'Current permissions policies (4)' and 'Other permissions policies (1/1162)'. In the 'Other permissions policies' section, the policy 'AmazonEFSCSIDriverPolicy' is selected. The 'Attach permissions' button at the bottom right is highlighted with a red checkmark.

Amazon EFS

File systems

fs-00337e6deab6507e2

Elastic File System

File systems

Access points

AWS Backup

AWS DataSync

AWS Transfer

Documentation

Performance mode

General Purpose

Throughput mode

Elastic

Lifecycle management

Transition into Infrequent Access (IA): 30 day(s) since last access

Transition into Archive: 90 day(s) since last access

Transition into Standard: None

Availability zone

Regional

b3f9b160-102b-4b87-8405-b98206001c02 (aws/elasticfilesystem)

File system state

Available

DNS name

fs-00337e6deab6507e2.efs.us-east-1.amazonaws.com

Replication overwrite protection

Enabled

Metered size

Monitoring

Tags

File system policy

Access points

Network

Replication

Network

Manage

Availability zone (AZ-ID)	Mount target ID	Subnet ID	VPC ID	Mount target state	IPv4 address	IPv6 address	Network interface ID	Security groups
us-east-1a (use1-az1)	fsmt-0e50c21be38a28de7	subnet-05bc60bf2a8a4a287	vpc-0f0c4fccc00165b45	Available	192.168.30.207	N/A	eni-0c61f73fe90b7c078	sg-0a4ed7cc0b65df866 (default)
us-east-1f (use1-az5)	fsmt-05d612931051f7a4b	subnet-08f76629255689f24	vpc-0f0c4fccc00165b45	Available	192.168.112.91	N/A	eni-08b70b3c060e686a7	sg-0a4ed7cc0b65df866 (default)

EC2

Security Groups

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Security Groups (1)

Find security groups by attribute or tag

sg-0a4ed7cc0b65df866

Clear filters

Actions

Export security groups to CSV

Create security group

Name	Security group ID	Security group name	VPC ID	Description
-	sg-0a4ed7cc0b65df866	default	vpc-0f0c4fccc00165b45	default VPC security group

EC2

Security Groups

sg-0a4ed7cc0b65df866 - default

Launch Templates

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AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

sg-0a4ed7cc0b65df866 - default

Details

Security group name

default

Security group ID

sg-0a4ed7cc0b65df866

Description

default VPC security group

VPC ID

vpc-0f0c4fccc00165b45

Owner

459532536509

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-072970baacbcd655	-	All traffic	All	All	sg-0a4ed7cc0b65df866...	-

EC2

Instances

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Instances (1/3)

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Connect

Instance state

Actions

Launch instances

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
efsc1uster-b17dng-Node	i-089a5ce9cab3d4a70	Running	m7i-flex.large	3/3 checks passed	us-east-1a	ec2-44-204-165-232.co...	44.204.165.232	-
efsc1uster-b17dng-Node	i-0f360d17d7e0dcf92	Running	m7i-flex.large	3/3 checks passed	us-east-1a	ec2-44-213-99-30.co...	44.213.99.30	-
efsc1uster-b17dng-Node	i-0f2712848b506fd53	Running	m7i-flex.large	3/3 checks passed	us-east-1f	ec2-44-197-103-91.co...	44.197.103.91	-

i-089a5ce9cab3d4a70 (efsc1uster-b17dng-Node)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Security details

IAM role

eksctl-efsc1uster-nodegroup-b17dng-NodeInstanceRole-RQ0amXZ2liuQ

Owner ID

459532536509

Launch time

Sat Jun 27 2026 18:42:13 GMT+0530 (India Standard Time)

Security groups

sg-0a0bbb73d748b307d (eks-cluster-sg-efsc1uster-1382028744)

EC2 > Security Groups > sg-0a4ed7cc0b65df866 - default > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info
 Security group rule ID
 sgr-072970baacbcd655

Type Info
 All traffic

Protocol Info
 All

Port range Info
 All

Source Info
 Custom

Description - optional Info
 sg-0a4ed7cc0b65df866 X

Delete

NFS

TCP

2049

Custom

sg-0a0bb73d748b307d X

sg-0a0bb73d748b307d X

Delete

Add rule

Cancel
 Preview changes
 Save rules

EC2 > Security Groups > sg-0a4ed7cc0b65df866 - default

Inbound security group rules successfully modified on security group (sg-0a4ed7cc0b65df866 | default)

sg-0a4ed7cc0b65df866 - default

Details

Security group name default	Security group ID sg-0a4ed7cc0b65df866	Description default VPC security group	VPC ID vpc-0f0c4fccc00165b45
Owner 459532536509	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0f163224f1ad59fb4	-	NFS	TCP	2049	sg-0a0bb73d748b307d...	-
-	sgr-072970baacbcd655	-	All traffic	All	All	sg-0a4ed7cc0b65df866...	-

STATIC

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ cd practice/26_June-EFS/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS (main)
$ ll
total 4
drwxr-xr-x 1 Bhavana 197121 0 Jun 27 20:42 EFS-Dynamic/
drwxr-xr-x 1 Bhavana 197121 0 Jun 27 20:46 EFS-Static/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS (main)
$ cd EFS-Static/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ ll
total 6
-rw-r--r-- 1 Bhavana 197121 341 Jun 27 20:44 1-efs-pv.yaml
-rw-r--r-- 1 Bhavana 197121 234 Jun 27 20:46 2-efs-pvc.yaml
-rw-r--r-- 1 Bhavana 197121 606 Jun 27 20:48 3-pod-svc.yaml
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl apply -f 1-efs-pv.yaml
persistentvolume/efs-static-pv created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl get pv
```

NAME	CAPACITY	ACCESS MODES	RECLAIM POLICY	STATUS	CLAIM	STORAGECLASS
VOLUMEATTRIBUTESCLASS	REASON	AGE				
efs-static-pv	5Gi	RWO	Retain	Available		
<unset>		9s				

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl apply -f 2-efs-pvc.yaml
persistentvolumeclaim/efs-static-pvc created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

```
$ kubectl get pvc
NAME                                STATUS    VOLUME             CAPACITY   ACCESS MODES   STORAGECLASS
efs-static-pvc                     Bound    efs-static-pv      5Gi        RWO              <unset>
9s
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

```
$ kubectl get pv
NAME                                CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM
efs-static-pv                     5Gi        RWO              Retain            Bound    default/efs-static-pvc
<unset>
45s
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

```
$ kubectl apply -f 3-pod-svc.yaml
pod/nginx-pod-efs-static created
service/nginx-lb created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

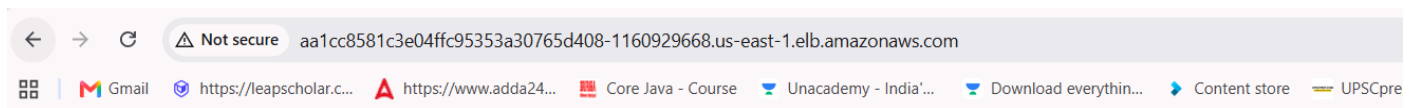
```
$ kubectl get all
NAME                                READY     STATUS    RESTARTS   AGE
pod/nginx-pod-efs-static           0/1       Pending   0           8s

NAME(S)          AGE      TYPE             CLUSTER-IP      EXTERNAL-IP
service/kubernetes 133m    ClusterIP        10.100.0.1      <none>
service/nginx-lb  8s      LoadBalancer    10.100.180.44   ab54753de4717416886829cab58923a1-976446243.us-east-1.elb.amazonaws.com 80:31032/TCP 8s
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

```
$ kubectl get all
NAME                                READY     STATUS    RESTARTS   AGE
pod/nginx-pod-efs-static           0/1       Pending   0           20s

NAME(S)          AGE      TYPE             CLUSTER-IP      EXTERNAL-IP
service/kubernetes 133m    ClusterIP        10.100.0.1      <none>
service/nginx-lb  20s     LoadBalancer    10.100.180.44   ab54753de4717416886829cab58923a1-976446243.us-east-1.elb.amazonaws.com 80:31032/TCP 20s
```

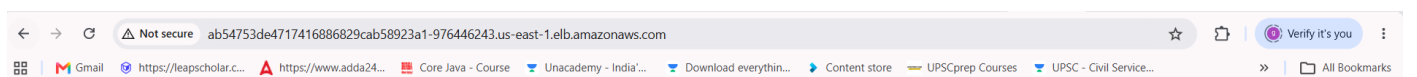


403 Forbidden

nginx/1.31.2

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

```
$ kubectl exec -it nginx-pod-efs-static -- bash
root@nginx-pod-efs-static:/# cd /usr/share/nginx/html/
root@nginx-pod-efs-static:/usr/share/nginx/html# cat index.html
cat: index.html: No such file or directory
root@nginx-pod-efs-static:/usr/share/nginx/html# echo "<h1> This Content is serving from EFS
static provisioning </h1>" > index.html
root@nginx-pod-efs-static:/usr/share/nginx/html# cat index.html
<h1> This Content is serving from EFS static provisioning </h1>
root@nginx-pod-efs-static:/usr/share/nginx/html# exit
exit
```



This Content is serving from EFS static provisioning

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl delete -f 3-pod-svc.yaml
pod "nginx-pod-efs-static" deleted from default namespace
service "nginx-lb" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl apply -f 3-pod-svc.yaml
pod/nginx-pod-efs-static2 created
service/nginx-lb created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/nginx-pod-efs-static2	1/1	Running	0	13s

NAME	PORT(S)	AGE	TYPE	CLUSTER-IP	EXTERNAL-IP
service/kubernetes	443/TCP	143m	ClusterIP	10.100.0.1	<none>
service/nginx-lb			LoadBalancer	10.100.125.245	a8646b53686d4410685836127e39f916-1569335605.us-east-1.elb.amazonaws.com



```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl delete -f 3-pod-svc.yaml
pod "nginx-pod-efs-static" deleted from default namespace
service "nginx-lb" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl apply -f 3-pod-svc.yaml
pod/nginx-pod-efs-static2 created
service/nginx-lb created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/nginx-pod-efs-static2	1/1	Running	0	13s

NAME	PORT(S)	AGE	TYPE	CLUSTER-IP	EXTERNAL-IP
service/kubernetes	443/TCP	143m	ClusterIP	10.100.0.1	<none>
service/nginx-lb			LoadBalancer	10.100.125.245	a8646b53686d4410685836127e39f916-1569335605.us-east-1.elb.amazonaws.com

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl delete -f 1-efs-pv.yaml
persistentvolume "efs-static-pv" deleted
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
```

```
$ kubectl get pv
```

NAME	CAPACITY	ACCESS MODES	RECLAIM POLICY	STATUS	CLAIM
STORAGECLASS	VOLUMEATTRIBUTESCLASS	REASON	AGE		
efs-static-pv	5Gi	RWO	Retain	Terminating	default/efs-static-pvc
<unset>		17m			

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl delete -f 2-efs-pvc.yaml
persistentvolumeclaim "efs-static-pvc" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Static (main)
$ kubectl delete -f 3-pod-svc.yaml
pod "nginx-pod-efs-static2" deleted from default namespace
service "nginx-lb" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-static (main)
```

```
$ kubectl get all
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.100.0.1	<none>	443/TCP	156m

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-static (main)
```

```
$ kubectl delete -f 3-pod-svc.yaml
```

```
Error from server (NotFound): error when deleting "3-pod-svc.yaml": pods "nginx-pod-efs-static2" not found
```

```
Error from server (NotFound): error when deleting "3-pod-svc.yaml": services "nginx-lb" not found
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-static (main)
```

```
$ kubectl get pv
```

```
No resources found
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-static (main)
```

```
$ kubectl get pvc
```

```
No resources found in default namespace.
```

DYNAMIC

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-static (main)
```

```
$ cd ..
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS (main)
```

```
$ ls
```

```
efs-dynamic/  efs-static/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS (main)
```

```
$ cd efs-dynamic/
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ ll
```

```
total 2
```

```
-rw-r--r-- 1 Bhavana 197121 461 Jun 27 21:18 1-efs-dya-sc-pvc.yaml
```

```
-rw-r--r-- 1 Bhavana 197121 569 Jun 27 21:18 2-pod-svc.yaml
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ kubectl apply -f 1-efs-dya-sc-pvc.yaml
```

```
storageclass.storage.k8s.io/efs-dynamic-sc created
```

```
persistentvolumeclaim/efs-dynamic created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ kubectl get sv
```

```
error: the server doesn't have a resource type "sv"
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ kubectl get sc
```

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	
ALLOWVOLUMEEXPANSION	AGE			
efs-dynamic-sc	efs.csi.aws.com	Delete	Immediate	false
21s				
gp2	kubernetes.io/aws-efs	Delete	WaitForFirstConsumer	false
163m				

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ kubectl get pvc
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES
STORAGECLASS	VOLUMEATTRIBUTESCLASS	AGE		
efs-dynamic	Bound	pvc-c6c82247-d30a-4841-bfac-646e451b6c60	5Gi	RWO
dynamic-sc	<unset>	26s		efs-

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ kubectl apply -f 2-pod-svc.yaml
```

```
pod/pod-nginx-efs-dynamic created
```

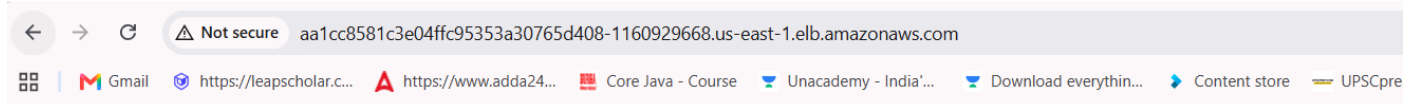
```
service/nginx-lb created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-dynamic (main)
```

```
$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/pod-nginx-efs-dynamic	1/1	Running	0	7s

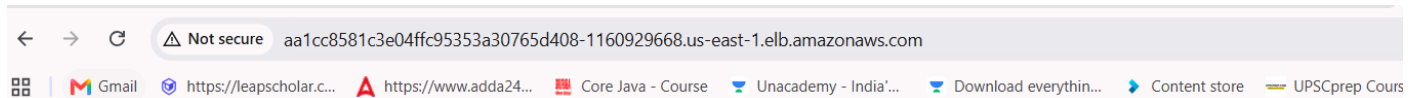
NAME	AGE	TYPE	CLUSTER-IP	EXTERNAL-IP
service/kubernetes	164m	ClusterIP	10.100.0.1	<none>
service/nginx-lb		LoadBalancer	10.100.212.44	aa1cc8581c3e04ffc95353a30765d408-1160929668.us-east-1.elb.amazonaws.com



403 Forbidden

nginx/1.31.2

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Dynamic (main)
$ kubectl exec -it pod-nginx-efs-dynamic -- bash
root@pod-nginx-efs-dynamic:/# cd /usr/share/nginx/html/
root@pod-nginx-efs-dynamic:/usr/share/nginx/html# cat index.html
cat: index.html: No such file or directory
root@pod-nginx-efs-dynamic:/usr/share/nginx/html# echo "<h1>This Content is serving from EFS Dynamic provisioning</h1>" > index.html
root@pod-nginx-efs-dynamic:/usr/share/nginx/html# cat index.html
<h1>This Content is serving from EFS Dynamic provisioning</h1>
root@pod-nginx-efs-dynamic:/usr/share/nginx/html# exit
exit
```



This Content is serving from EFS Dynamic provisioning

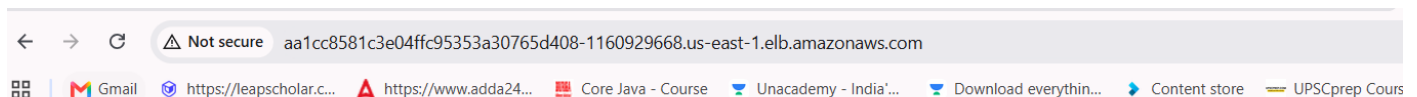
```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Dynamic (main)
$ kubectl delete po pod-nginx-efs-dynamic
pod "pod-nginx-efs-dynamic" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Dynamic (main)
$ kubectl apply -f 2-pod-svc.yaml
pod/pod-nginx-efs-dynamic2 created
service/nginx-lb unchanged
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/EFS-Dynamic (main)
$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/pod-nginx-efs-dynamic2	1/1	Running	0	8s

NAME	AGE	TYPE	CLUSTER-IP	EXTERNAL-IP
service/kubernetes	169m	ClusterIP	10.100.0.1	<none>
service/nginx-lb		LoadBalancer	10.100.212.44	aa1cc8581c3e04ffc95353a30765d408-1160929668.us-east-1.elb.amazonaws.com



This Content is serving from EFS Dynamic provisioning

Amazon EFS > File systems > fs-00337e6deab6507e2

Elastic File System

File systems
Access points
AWS Backup
AWS DataSync
AWS Transfer
Documentation

General

Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-1:459532536509:file-system/fs-00337e6deab6507e2

Performance mode
General Purpose

Throughput mode
Elastic

Lifecycle management
Transition into Infrequent Access (IA): 30 day(s) since last access
Transition into Archive: 90 day(s) since last access
Transition into Standard: None

Availability zone
Regional

Automatic backups
Enabled

Encrypted
b3f9b160-102b-4b87-8405-b98206001c02 (aws/elasticfilesystem)

File system state
Available

DNS name
fs-00337e6deab6507e2.efs.us-east-1.amazonaws.com

Replication overwrite protection
Enabled

Metered size | Monitoring | Tags | File system policy | **Access points** | Network | Replication

Access points (1)

Search access points by name or ID

Name	Access point ID	Path	POSIX user	Creation info	State
z	fsap-05c985eb96640f268	/prod/pvc-c6c82247-d30a-48...	50000 : 50000	50000 : 50000 (700)	Available

Amazon EFS > Access points > fsap-05c985eb96640f268

Elastic File System

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Documentation

Details

File system ID
fs-00337e6deab6507e2

Root directory path
/prod/pvc-c6c82247-d30a-4841-bfac-646e451b6c60

State
Available

Access point ARN
arn:aws:elasticfilesystem:us-east-1:459532536509:access-point/fsap-05c985eb96640f268

POSIX user

User ID
50000

Group ID
50000

Secondary group IDs
-

Root directory creation permissions

Owner user ID
50000

Owner group ID
50000

Permissions
700

Delete Attach

EC2 > Security Groups > sg-0a4ed7cc0b65df866 - default

sg-0a4ed7cc0b65df866 - default

Actions

Details

Security group name
default

Security group ID
sg-0a4ed7cc0b65df866

Description
default VPC security group

VPC ID
vpc-0f0c8fccc00165b45

Owner
459532536509

Inbound rules count
2 Permission entries

Outbound rules count
1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-0f163224f1ad59fb4	-	NFS	TCP	2049	sg-0a0bb73d748b307...	-
-	sg-072970baacbd6e55	-	All traffic	All	All	sg-0a4ed7cc0b65df866...	-

remove

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/26_June-EFS/efs-Dynamic (main)

```
$ eksctl delete cluster efsc1uster
```